PREMKUMAR.P

Aspiring Mechanical Engineer

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CAREER OBJECTIVE

Final-year Mechanical Engineering student with hands-on project experience in CAD modelling, structural simulation, and topology optimization. Seeking an internship opportunity to apply strong fundamentals in design, analysis, and manufacturing while contributing to real-world engineering challenges and continuing to build practical, industry-relevant skills.

EDUCATION

Bachelor of Engineering (B.E.) – Mechanical Engineering | Expected 2027
Government College of Engineering, Salem

TECHNICAL PROJECTS

Robotic Arm Design via Topology Optimization

- Designed a robotic arm structure in CATIA/SolidWorks and applied topology optimization techniques to identify and remove non-load-bearing material.
- Achieved significant weight reduction while preserving structural strength and stiffness, validated through simulation in Ansys Workbench.
- Analyzed stress distribution and deformation under working loads to ensure the optimized design met safety and performance requirements.

Smart Solar-Powered Material Handling Conveyor

- Designed and developed a complete solar-powered conveyor system for material handling, from CAD modelling to a working physical prototype.
- Performed simulation and load analysis to size components and validate mechanical performance before fabrication.
- Integrated a solar power source to drive the conveyor mechanism, demonstrating an energy-efficient, low-cost material handling solution suitable for small-scale industrial use.

TECHNICAL SKILLS

Design & Modelling	CATIA, SolidWorks
Simulation & Analysis	Ansys Workbench (Structural & Stress Analysis)
Core Mechanical Concepts	Strength of Materials, Thermodynamics, Machine Design
Methodologies	Six Sigma fundamentals, Topology Optimization, Engineering Drawing & GD&T;

CERTIFICATIONS

- Six Sigma – LinkedIn Learning
- CATIA and SolidWorks – Coursera

KEY STRENGTHS

- Strong foundation in mechanical design principles with practical application through independent project work.
- Proficient in translating conceptual designs into validated, simulation-tested engineering models.
- Quick learner, detail-oriented, and committed to continuous upskilling through self-driven certifications.
- Effective at working independently to carry a project from concept through prototype completion.
- Comfortable collaborating in team environments and communicating technical ideas clearly.
- Genuine interest in robotics, automation, and renewable energy applications within mechanical systems.